

NEWTON'S INTERPOLATION FORMULA AND SUMS OF POWERS

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ABSTRACT. In this manuscript, we derive formulas for multifold sums of powers by utilizing Newton's interpolation formula. Furthermore, we provide formulas for multifold sums of powers in terms of Stirling numbers of the second kind and Eulerian numbers.

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1. INTRODUCTION

2. AUXILIARY LEMMAS

For multifold sums of powers, we use the notation provided by Donald Knuth in [?], because it is simple and beautiful.

Definition 2.1 (Multifold sums of powers). *For non-negative integers r, n, m*

$$\Sigma^0 n^m = n^m,$$

$$\Sigma^1 n^m = \Sigma^0 1^m + \Sigma^0 2^m + \cdots + \Sigma^0 n^m,$$

$$\Sigma^{r+1} n^m = \Sigma^r 1^m + \Sigma^r 2^m + \cdots + \Sigma^r n^m.$$

We utilize the following notation for rising factorials

Definition 2.2 (Rising factorials). *For integers n, k*

$$n^{(k)} = \begin{cases} 0, & \text{if } k < 0, \\ 1, & \text{if } k = 0, \\ \prod_{j=0}^{k-1} (n + j), & \text{if } k > 0. \end{cases}$$

We encourage the audience to read J. F. Steffensen's work on the definition of generalized factorials [?].

Lemma 2.3 (Forward Newton's formula). *Let $f: \mathbb{Z} \rightarrow \mathbb{C}$ be a function, and let $t \in \mathbb{Z}$. Then, for all $n \in \mathbb{Z}$,*

$$f(n) = \sum_{k=0}^{\infty} \frac{(n-t)^{(k)}}{k!} \Delta^k f(t) = \sum_{k=0}^{\infty} \binom{n-t}{k} \Delta^k f(t),$$

where the k -th forward difference $\Delta^k f(t)$ of f evaluated at t is defined by

$$\Delta^k f(t) = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} f(t+j).$$

Proof. By [?, section 2, eq. (19)], and by identity $\frac{(n-t)^{(k)}}{k!} = \binom{n-t}{k}$. □

Lemma 2.4 (Forward Newton's formula for powers). *For non-negative integers n, m , and an arbitrary integer t*

$$n^m = \sum_{k=0}^m \binom{n-t}{k} \Delta^k t^m.$$

Proof. Special case of Lemma (2.3) for powers. □

Lemma 2.5 (Generalized hockey-stick identity). *For arbitrary integers a, b and j ,*

$$\sum_{k=a}^b \binom{k}{j} = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

Proof. We have, $\sum_{k=a}^b \binom{k}{j} = \binom{a}{j} + \binom{a+1}{j} + \cdots + \binom{b}{j} = \left(\sum_{k=0}^b \binom{k}{j} \right) - \left(\sum_{k=0}^{a-1} \binom{k}{j} \right)$. By hockey-stick identity $\sum_{k=0}^n \binom{k}{j} = \binom{n+1}{j+1}$ yields,

$$\sum_{k=a}^b \binom{k}{j} = \left(\sum_{k=0}^b \binom{k}{j} \right) - \left(\sum_{k=0}^{a-1} \binom{k}{j} \right) = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

This completes the proof. □

Lemma 2.6 (Forward hockey-stick identity). *For integer $n \geq 1$, and arbitrary integers t, k, r*

$$\sum_{j=1}^n \binom{j-t+r}{k+r} = \binom{n-t+r+1}{k+r+1} - \binom{1-t+r}{k+r+1}.$$

Proof. By setting $a = 1 - t + r$, and $b = n - t + 1$ into Lemma (2.5). □

Lemma 2.7 (Multifold sums of zero powers). *For non-negative integers r, n*

$$\Sigma^r n^0 = \binom{r+n-1}{r}.$$

Proof.

- (1) Let $r = 0$, then we have $\Sigma^0 n^0 = 1$, by definition (2.1).
- (2) Let $r = 1$, then we have $\Sigma^1 n^0 = \sum_{k=1}^n 1 = \binom{n}{1}$.
- (3) Let $r = 2$, then we have $\Sigma^2 n^0 = \sum_{k=1}^n \binom{k}{1} = \binom{n+1}{2}$.
- (4) Let $r = 3$, then we have $\Sigma^3 n^0 = \sum_{k=1}^n \binom{k+1}{2} = \binom{n+2}{3}$.
- (5) Similarly, for arbitrary integer r , by hockey-stick identity $\sum_{k=1}^n \binom{k}{r} = \binom{n+1}{r+1}$, yields

$$\Sigma^r n^0 = \sum_{k=1}^n \Sigma^{r-1} k^0 = \sum_{k=1}^n \binom{k+r-2}{r-1} = \binom{r+n-1}{r}.$$

This completes the proof. $\square$

Convention 2.8. *For all x*

$$x^0 = 1.$$

Donald Knuth give extensive discussion on the convention $x^0 = 1$ for all x , including zero, in Concrete Mathematics [?, p. 162], and in [?].

3. MAIN RESULTS

We start our discussion from Lemma (??), which yields forward Newton's interpolation formula for powers. For example,

$$\begin{aligned} n^3 &= 0 \binom{n-0}{0} + 1 \binom{n-0}{1} + 6 \binom{n-0}{2} + 6 \binom{n-0}{3}, \\ n^3 &= 1 \binom{n-1}{0} + 7 \binom{n-1}{1} + 12 \binom{n-1}{2} + 6 \binom{n-1}{3}, \\ n^3 &= 8 \binom{n-2}{0} + 19 \binom{n-2}{1} + 18 \binom{n-2}{2} + 6 \binom{n-2}{3}, \\ n^3 &= 27 \binom{n-3}{0} + 37 \binom{n-3}{1} + 24 \binom{n-3}{2} + 6 \binom{n-3}{3}. \end{aligned}$$

Thus, the transition to formula for sums of powers is quite straightforward,

$$\Sigma^1 n^m = \sum_{k=0}^m \left[\sum_{j=1}^n \binom{j-t}{k} \right] \Delta^k t^m.$$

Therefore, by setting $r = 0$ into Lemma (2.6) yields closed form of ordinary sums of powers

Proposition 3.1 (Ordinary sums of powers). *For non-negative integers n, m , and an arbitrary integer t*

$$\Sigma^1 n^m = \sum_{k=0}^m \left[\binom{n-t+1}{k+1} - \binom{1-t}{k+1} \right] \Delta^k t^m.$$

We may see that the basis $\binom{j-t+r}{k+r}$ for repeatedly applied Lemma (2.6) was indeed chosen correctly, because we derived closed form for sums of powers with ease, simply setting parameter for r . For example,

Example 3.2. For integer $0 \leq t \leq 3$, we have

$$\begin{aligned}\Sigma^1 n^3 &= 0\left[\binom{n+1}{1} - \binom{1}{1}\right] + 1\left[\binom{n+1}{2} - \binom{1}{2}\right] + 6\left[\binom{n+1}{3} - \binom{1}{3}\right] + 6\left[\binom{n+1}{4} - \binom{1}{4}\right], \\ \Sigma^1 n^3 &= 1\left[\binom{n}{2} - \binom{0}{2}\right] + 7\left[\binom{n}{3} - \binom{0}{3}\right] + 12\left[\binom{n}{4} - \binom{0}{4}\right] + 6\left[\binom{n}{5} - \binom{0}{5}\right], \\ \Sigma^1 n^3 &= 8\left[\binom{n-1}{2} - \binom{-1}{2}\right] + 19\left[\binom{n-1}{3} - \binom{-1}{3}\right] + 18\left[\binom{n-1}{4} - \binom{-1}{4}\right] + 6\left[\binom{n-1}{5} - \binom{-1}{5}\right], \\ \Sigma^1 n^3 &= 27\left[\binom{n-2}{2} - \binom{-2}{2}\right] + 37\left[\binom{n-2}{3} - \binom{-2}{3}\right] + 24\left[\binom{n-2}{4} - \binom{-2}{4}\right] + 6\left[\binom{n-2}{5} - \binom{-2}{5}\right].\end{aligned}$$

From example above, we may observe binomial coefficients of negative upper index, which may look unfamiliar. However, there is no reason to be afraid of, because negative upper index binomial coefficients are completely deterministic, because $\binom{-n}{k} = (-1)^k \binom{k+n-1}{k}$. By setting $t = 0$ into Proposition (3.1) yields one more well-known identity for sums of powers, mentioned in [?, p. 190], and [?]

Corollary 3.3. For non-negative integers n, m

$$\Sigma^1 n^m = \sum_{j=0}^m \binom{n+1}{j+1} \Delta^j 0^m.$$

For example,

Example 3.4. For non-negative integer n , we have

$$\begin{aligned}\Sigma^1 n^2 &= 0\binom{n+1}{1} + 1\binom{n+1}{2} + 2\binom{n+1}{3}, \\ \Sigma^1 n^3 &= 0\binom{n+1}{1} + 1\binom{n+1}{2} + 6\binom{n+1}{3} + 6\binom{n+1}{4}, \\ \Sigma^1 n^4 &= 0\binom{n+1}{1} + 1\binom{n+1}{2} + 14\binom{n+1}{3} + 36\binom{n+1}{4} + 24\binom{n+1}{5}, \\ \Sigma^1 n^5 &= 0\binom{n+1}{1} + 1\binom{n+1}{2} + 30\binom{n+1}{3} + 150\binom{n+1}{4} + 240\binom{n+1}{5} + 120\binom{n+1}{6}.\end{aligned}$$

From the example above, we observe the coefficients $0, 1, 2, 0, 1, 6, 6, 0, 1, 14, 36, 24, \dots$, which is the sequence [A131689](#) in the OEIS [?].

4. STIRLING FORM

5. EULERIAN FORM

6. BACKWARD DIFFERENCE FORM

7. CENTRAL DIFFERENCE FORM

8. CONCLUSIONS

9. ACKNOWLEDGEMENTS

10. MATHEMATICA

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